

Solving the Basic New Keynesian Model using the Blanchard-Kahn Method

The three-equation model is:¹

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t \quad \text{DAS}$$

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\sigma} (\tilde{i}_t - \mathbb{E}_t \pi_{t+1} - \tilde{r}_t^n) \quad \text{DIS}$$

$$\tilde{i}_t = \phi_\pi \pi_t + \phi_y \tilde{y}_t + v_t \quad \text{TR}$$

where $\tilde{A}_{t+1} = \rho_a \tilde{A}_t + \varepsilon_{t+1}^A$, $v_{t+1} = \rho_v v_t + \varepsilon_{t+1}^v$ and $\tilde{r}_t^n = -\sigma \psi_y^n (1 - \rho_a) \tilde{A}_t$.²

Substituting TR into DIS and rearranging we obtain the following two-equation model:

$$\beta \mathbb{E}_t \pi_{t+1} = \pi_t - \kappa \tilde{y}_t \quad \text{DAS}$$

$$\mathbb{E}_t \pi_{t+1} + \sigma \mathbb{E}_t \tilde{y}_{t+1} = \phi_\pi \pi_t + (\sigma + \phi_y) \tilde{y}_t - \tilde{r}_t^n + v_t \quad \text{DAD}$$

Then:

$$\begin{bmatrix} \beta & 0 \\ 1 & \sigma \end{bmatrix} \begin{bmatrix} \mathbb{E}_t \pi_{t+1} \\ \mathbb{E}_t \tilde{y}_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & -\kappa \\ \phi_\pi & \sigma + \phi_y \end{bmatrix} \begin{bmatrix} \pi_t \\ \tilde{y}_t \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} \tilde{r}_t^n \\ v_t \end{bmatrix}$$

or:

$$A \mathbb{E}_t w_{t+1} = B w_t + C \eta_t$$

where

$$A \equiv \begin{bmatrix} \beta & 0 \\ 1 & \sigma \end{bmatrix}, \quad B \equiv \begin{bmatrix} 1 & -\kappa \\ \phi_\pi & \sigma + \phi_y \end{bmatrix}, \quad C \equiv \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix}, \quad w_t \equiv \begin{bmatrix} \pi_t \\ \tilde{y}_t \end{bmatrix}, \quad \eta_t \equiv \begin{bmatrix} \tilde{r}_t^n \\ v_t \end{bmatrix}$$

Since A is nonsingular, we can rewrite the expression above in the form of Blanchard & Kahn (1980). Premultiplying both sides by A^{-1} we get: $\mathbb{E}_t w_{t+1} = A^{-1} B w_t + A^{-1} C \eta_t \Rightarrow$

$$\mathbb{E}_t w_{t+1} = \Omega w_t + \Theta \eta_t$$

where

$$\begin{aligned} \Omega &\equiv A^{-1} B = \begin{bmatrix} \frac{1}{\beta} & 0 \\ -\frac{1}{\sigma\beta} & \frac{1}{\sigma} \end{bmatrix} \begin{bmatrix} 1 & -\kappa \\ \phi_\pi & \sigma + \phi_y \end{bmatrix} = \begin{bmatrix} \frac{1}{\beta} & -\frac{\kappa}{\beta} \\ \frac{\beta\phi_\pi - 1}{\sigma\beta} & \frac{\beta(\sigma + \phi_y) + \kappa}{\sigma\beta} \end{bmatrix} \\ \Theta &\equiv A^{-1} C = \begin{bmatrix} \frac{1}{\beta} & 0 \\ -\frac{1}{\sigma\beta} & \frac{1}{\sigma} \end{bmatrix} \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -\frac{1}{\sigma} & \frac{1}{\sigma} \end{bmatrix} \end{aligned}$$

Now we want to diagonalize Ω . That is, we want to write

$$\Omega = \Gamma \Lambda \Gamma^{-1}$$

where Λ is a diagonal matrix with the eigenvalues of Ω in its diagonal, and Γ is the matrix having as columns the right-eigenvectors of Ω (see the Appendix).³

Using the expression above into $\mathbb{E}_t w_{t+1} = \Omega w_t + \Theta \eta_t$ we get:

$$\mathbb{E}_t w_{t+1} = \Gamma \Lambda \Gamma^{-1} w_t + \Theta \eta_t$$

Premultiplying by Γ^{-1} :

$$\Gamma^{-1} \mathbb{E}_t w_{t+1} = \Lambda \Gamma^{-1} w_t + \Gamma^{-1} \Theta \eta_t$$

Then:

$$\mathbb{E}_t (\Gamma^{-1} w_{t+1}) = \Lambda \Gamma^{-1} w_t + \Gamma^{-1} \Theta \eta_t$$

¹ See Chapter 3 in Galí (2008).

² Notice that $\tilde{r}_{t+1}^n = -\sigma \psi_y^n (1 - \rho_a) \tilde{A}_{t+1} \Rightarrow \tilde{r}_{t+1}^n = -\sigma \psi_y^n (1 - \rho_a) (\rho_a \tilde{A}_t + \varepsilon_{t+1}^A) \Rightarrow$

$\tilde{r}_{t+1}^n = -\sigma \psi_y^n (1 - \rho_a) \rho_a \tilde{A}_t - \sigma \psi_y^n (1 - \rho_a) \varepsilon_{t+1}^A \Rightarrow \tilde{r}_{t+1}^n = \rho_a \tilde{r}_t^n + \varepsilon_{t+1}^r$, where $\varepsilon_{t+1}^r \equiv -\sigma \psi_y^n (1 - \rho_a) \varepsilon_{t+1}^A$.

³ Instead of the standard diagonalization method, Blanchard & Kahn use the more general Jordan (canonical) decomposition (which applies also to non-diagonalizable matrices). See Chapter 21 in Sydsæter, Strøm & Berck (2005) for a concise exposition.

Then:

$$\mathbb{E}_t z_{t+1} = \Lambda z_t + \Phi \eta_t$$

where $z_t \equiv \Gamma^{-1} w_t$ and $\Phi \equiv \Gamma^{-1} \Theta$. Notice that each element of z_t is a linear combination of the elements of w_t (π_t and $\tilde{y}_t$).

The expression above gives:

$$\begin{bmatrix} \mathbb{E}_t z_{1,t+1} \\ \mathbb{E}_t z_{2,t+1} \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} z_{1,t} \\ z_{2,t} \end{bmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} \tilde{r}_t^n \\ v_t \end{bmatrix}$$

Then:

$$\mathbb{E}_t z_{1,t+1} = \lambda_1 z_{1,t} + \phi_{11} \tilde{r}_t^n + \phi_{12} v_t \quad (1)$$

$$\mathbb{E}_t z_{2,t+1} = \lambda_2 z_{2,t} + \phi_{21} \tilde{r}_t^n + \phi_{22} v_t \quad (2)$$

Notice that the diagonalization of Ω has allowed us to transform the original system of equations into two separate equations that can be solved individually.

Now we focus on (1).⁴ We first notice that we don't have an initial (boundary) condition for $z_{1,t}$.⁵ Instead, our boundary condition will be a no-explosion condition to be specified below. If $\lambda_1 > 1$, there will be a unique non-explosive solution that can be found by *forward* iteration. To find this solution we start by rewriting (1) as follows:⁶

$$z_{1,t} = -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) + \frac{1}{\lambda_1} \mathbb{E}_t z_{1,t+1}$$

Then:

$$\begin{aligned} z_{1,t} &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) + \frac{1}{\lambda_1} \mathbb{E}_t z_{1,t+1} \\ &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) + \frac{1}{\lambda_1} \mathbb{E}_t \left\{ -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_{t+1}^n + \phi_{12} v_{t+1}) + \frac{1}{\lambda_1} \mathbb{E}_{t+1} z_{1,t+2} \right\} \\ &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) - \left(\frac{1}{\lambda_1} \right)^2 \mathbb{E}_t \{ \phi_{11} \tilde{r}_{t+1}^n + \phi_{12} v_{t+1} \} + \left(\frac{1}{\lambda_1} \right)^2 \mathbb{E}_t \{ \mathbb{E}_{t+1} z_{1,t+2} \} \\ &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) - \left(\frac{1}{\lambda_1} \right)^2 (\phi_{11} \rho_a \tilde{r}_t^n + \phi_{12} \rho_v v_t) + \left(\frac{1}{\lambda_1} \right)^2 \mathbb{E}_t z_{1,t+2} \\ &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) - \left(\frac{1}{\lambda_1} \right)^2 (\phi_{11} \rho_a \tilde{r}_t^n + \phi_{12} \rho_v v_t) + \left(\frac{1}{\lambda_1} \right)^2 \mathbb{E}_t \left\{ -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_{t+2}^n + \phi_{12} v_{t+2}) + \frac{1}{\lambda_1} \mathbb{E}_{t+2} z_{1,t+3} \right\} \\ &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) - \left(\frac{1}{\lambda_1} \right)^2 (\phi_{11} \rho_a \tilde{r}_t^n + \phi_{12} \rho_v v_t) - \left(\frac{1}{\lambda_1} \right)^3 \mathbb{E}_t \{ \phi_{11} \tilde{r}_{t+2}^n + \phi_{12} v_{t+2} \} + \left(\frac{1}{\lambda_1} \right)^3 \mathbb{E}_t \{ \mathbb{E}_{t+2} z_{1,t+3} \} \\ &= -\frac{1}{\lambda_1} (\phi_{11} \tilde{r}_t^n + \phi_{12} v_t) - \left(\frac{1}{\lambda_1} \right)^2 (\phi_{11} \rho_a \tilde{r}_t^n + \phi_{12} \rho_v v_t) - \left(\frac{1}{\lambda_1} \right)^3 (\phi_{11} \rho_a^2 \tilde{r}_t^n + \phi_{12} \rho_v^2 v_t) + \left(\frac{1}{\lambda_1} \right)^3 \mathbb{E}_t z_{1,t+3} \\ &= -\frac{\phi_{11}}{\lambda_1} \left[1 + \frac{\rho_a}{\lambda_1} + \left(\frac{\rho_a}{\lambda_1} \right)^2 \right] \tilde{r}_t^n - \frac{\phi_{12}}{\lambda_1} \left[1 + \frac{\rho_v}{\lambda_1} + \left(\frac{\rho_v}{\lambda_1} \right)^2 \right] v_t + \left(\frac{1}{\lambda_1} \right)^3 \mathbb{E}_t z_{1,t+3} \\ &\dots \\ &= -\frac{\phi_{11}}{\lambda_1} \left[1 + \frac{\rho_a}{\lambda_1} + \left(\frac{\rho_a}{\lambda_1} \right)^2 + \dots + \left(\frac{\rho_a}{\lambda_1} \right)^{j-1} \right] \tilde{r}_t^n - \frac{\phi_{12}}{\lambda_1} \left[1 + \frac{\rho_v}{\lambda_1} + \left(\frac{\rho_v}{\lambda_1} \right)^2 + \dots + \left(\frac{\rho_v}{\lambda_1} \right)^{j-1} \right] v_t + \left(\frac{1}{\lambda_1} \right)^j \mathbb{E}_t z_{1,t+j} \end{aligned}$$

Then:

$$z_{1,t} = -\frac{\phi_{11}}{\lambda_1} \frac{1 - \left(\frac{\rho_a}{\lambda_1} \right)^j}{1 - \frac{\rho_a}{\lambda_1}} \tilde{r}_t^n - \frac{\phi_{12}}{\lambda_1} \frac{1 - \left(\frac{\rho_v}{\lambda_1} \right)^j}{1 - \frac{\rho_v}{\lambda_1}} v_t + \left(\frac{1}{\lambda_1} \right)^j \mathbb{E}_t z_{1,t+j}$$

For $j \rightarrow \infty$ we get:

$$z_{1,t} = -\frac{\phi_{11}}{\lambda_1} \frac{1}{1 - \frac{\rho_a}{\lambda_1}} \tilde{r}_t^n - \frac{\phi_{12}}{\lambda_1} \frac{1}{1 - \frac{\rho_v}{\lambda_1}} v_t + \lim_{j \rightarrow \infty} \left(\frac{1}{\lambda_1} \right)^j \mathbb{E}_t z_{1,t+j}$$

⁴The analysis for (2) is completely analogous.

⁵We know $z_{1,t}$ is a linear combination of π_t and $\tilde{y}_t$. But π_t and $\tilde{y}_t$ are nonpredetermined (jump) variables for which we don't have initial conditions.

⁶Notice that $\lambda_1 > 1 \Rightarrow \frac{1}{\lambda_1} < 1$.

Now we impose the following no-explosion condition:

$$\lim_{j \rightarrow \infty} \left(\frac{1}{\lambda_1} \right)^j \mathbb{E}_t z_{1,t+j} = 0$$

Then:

$$z_{1,t} = -\frac{\phi_{11}}{\lambda_1} \frac{\lambda_1}{\lambda_1 - \rho_a} \hat{r}_t^n - \frac{\phi_{12}}{\lambda_1} \frac{\lambda_1}{\lambda_1 - \rho_v} v_t$$

Finally:

$$z_{1,t} = \frac{\phi_{11}}{\rho_a - \lambda_1} \hat{r}_t^n + \frac{\phi_{12}}{\rho_v - \lambda_1} v_t \quad (3)$$

which is the *unique* non-explosive solution for $z_{1,t}$.

Analogously, the solution for $z_{2,t}$ is:

$$z_{2,t} = \frac{\phi_{21}}{\rho_a - \lambda_2} \hat{r}_t^n + \frac{\phi_{22}}{\rho_v - \lambda_2} v_t \quad (4)$$

Now we want to uncover the solutions for the original variables. From $z_t \equiv \Gamma^{-1} w_t$ we get $w_t = \Gamma z_t$. Then:

$$\begin{bmatrix} \pi_t \\ \tilde{y}_t \end{bmatrix} = \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} \begin{bmatrix} z_{1,t} \\ z_{2,t} \end{bmatrix}$$

where the elements of Γ have been denoted by γ_{ij} .

Then:

$$\pi_t = \gamma_{11} z_{1,t} + \gamma_{12} z_{2,t} \quad (5)$$

$$\tilde{y}_t = \gamma_{21} z_{1,t} + \gamma_{22} z_{2,t} \quad (6)$$

Substituting (3) and (4) into (5) and (6) we get:

$$\begin{aligned} \pi_t &= \gamma_{11} \left(\frac{\phi_{11}}{\rho_a - \lambda_1} \hat{r}_t^n + \frac{\phi_{12}}{\rho_v - \lambda_1} v_t \right) + \gamma_{12} \left(\frac{\phi_{21}}{\rho_a - \lambda_2} \hat{r}_t^n + \frac{\phi_{22}}{\rho_v - \lambda_2} v_t \right) \\ \tilde{y}_t &= \gamma_{21} \left(\frac{\phi_{11}}{\rho_a - \lambda_1} \hat{r}_t^n + \frac{\phi_{12}}{\rho_v - \lambda_1} v_t \right) + \gamma_{22} \left(\frac{\phi_{21}}{\rho_a - \lambda_2} \hat{r}_t^n + \frac{\phi_{22}}{\rho_v - \lambda_2} v_t \right) \end{aligned}$$

Then:

$$\begin{aligned} \pi_t &= \left(\frac{\gamma_{11}\phi_{11}}{\rho_a - \lambda_1} + \frac{\gamma_{12}\phi_{21}}{\rho_a - \lambda_2} \right) \hat{r}_t^n + \left(\frac{\gamma_{11}\phi_{12}}{\rho_v - \lambda_1} + \frac{\gamma_{12}\phi_{22}}{\rho_v - \lambda_2} \right) v_t \\ \tilde{y}_t &= \left(\frac{\gamma_{21}\phi_{11}}{\rho_a - \lambda_1} + \frac{\gamma_{22}\phi_{21}}{\rho_a - \lambda_2} \right) \hat{r}_t^n + \left(\frac{\gamma_{21}\phi_{12}}{\rho_v - \lambda_1} + \frac{\gamma_{22}\phi_{22}}{\rho_v - \lambda_2} \right) v_t \end{aligned}$$

Then:

$$\pi_t = \psi_{\pi r} \hat{r}_t^n + \psi_{\pi v} v_t \quad (7)$$

$$\tilde{y}_t = \psi_{y r} \hat{r}_t^n + \psi_{y v} v_t \quad (8)$$

where

$$\begin{aligned} \psi_{\pi r} &\equiv \frac{\gamma_{11}\phi_{11}}{\rho_a - \lambda_1} + \frac{\gamma_{12}\phi_{21}}{\rho_a - \lambda_2} \\ \psi_{\pi v} &\equiv \frac{\gamma_{11}\phi_{12}}{\rho_v - \lambda_1} + \frac{\gamma_{12}\phi_{22}}{\rho_v - \lambda_2} \\ \psi_{y r} &\equiv \frac{\gamma_{21}\phi_{11}}{\rho_a - \lambda_1} + \frac{\gamma_{22}\phi_{21}}{\rho_a - \lambda_2} \\ \psi_{y v} &\equiv \frac{\gamma_{21}\phi_{12}}{\rho_v - \lambda_1} + \frac{\gamma_{22}\phi_{22}}{\rho_v - \lambda_2} \end{aligned}$$

Using the fact that $\tilde{r}_t^n = -\sigma\psi_y^n(1 - \rho_a)\tilde{A}_t$ we can rewrite (7) and (8) as follows:

$$\pi_t = \psi_{\pi a}\tilde{A}_t + \psi_{\pi v}v_t \quad (9)$$

$$\tilde{y}_t = \psi_{ya}\tilde{A}_t + \psi_{yv}v_t \quad (10)$$

where

$$\psi_{\pi a} \equiv -\psi_{\pi r}\sigma\psi_y^n(1 - \rho_a)$$

$$\psi_{ya} \equiv -\psi_{yr}\sigma\psi_y^n(1 - \rho_a)$$

Using the Matlab code `nlm_bk.m` you can check that this solution coincides with the one we found in class using the method of undetermined coefficients.

Appendix

Recall that an eigenvalue of a square matrix Ω is a (possibly complex) number λ for which there is a (possibly complex) non-zero vector $\mathbf{v}$ such that $\Omega\mathbf{v} = \lambda\mathbf{v}$.⁷ The vector $\mathbf{v}$ is called a right eigenvector of Ω . From $\Omega\mathbf{v} = \lambda\mathbf{v}$ we get $(\Omega - \lambda I)\mathbf{v} = \mathbf{0}$. Therefore, to compute the eigenvalues of Ω we need to solve the equation $\det(\Omega - \lambda I) = 0$.

For $\Omega = \begin{bmatrix} \frac{1}{\beta} & -\frac{\kappa}{\beta} \\ \frac{\beta\phi_\pi - 1}{\beta\sigma} & \frac{\beta(\sigma + \phi_y) + \kappa}{\beta\sigma} \end{bmatrix}$, we need to solve:

$$\det \left(\begin{bmatrix} \frac{1}{\beta} - \lambda & -\frac{\kappa}{\beta} \\ \frac{\beta\phi_\pi - 1}{\beta\sigma} & \frac{\beta(\sigma + \phi_y) + \kappa}{\beta\sigma} - \lambda \end{bmatrix} \right) = 0$$

Then:

$$\lambda^2 - \left(\frac{1}{\beta} + \frac{\beta(\sigma + \phi_y) + \kappa}{\beta\sigma} \right) \lambda + \left(\frac{\beta(\sigma + \phi_y) + \kappa}{\beta^2\sigma} + \frac{\kappa(\beta\phi_\pi - 1)}{\beta^2\sigma} \right) = 0$$

$$\lambda^2 - \frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma} \lambda + \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma} = 0$$

where the left-hand-side is the characteristic polynomial of Ω .

Then, the roots of the characteristic polynomial are:

$$\lambda_{1,2} = \frac{1}{2} \left(\frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma} \pm \sqrt{\left(\frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma} \right)^2 - 4 \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma}} \right)$$

Then:

$$\lambda_{1,2} = \frac{1}{2} \left(\frac{\sigma + \beta\sigma + \beta\phi_y + \kappa}{\beta\sigma} \pm \sqrt{\frac{(\sigma + \beta\sigma + \beta\phi_y + \kappa)^2}{\beta^2\sigma^2} - 4 \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma}} \right)$$

$$\lambda_{1,2} = \frac{1}{2} \left(\frac{\sigma + \beta\sigma + \beta\phi_y + \kappa}{\beta\sigma} \pm \sqrt{\frac{(\sigma + \beta\sigma + \beta\phi_y + \kappa)^2 - 4(\sigma + \phi_y + \kappa\phi_\pi)\beta\sigma}{\beta^2\sigma^2}} \right)$$

$$\lambda_{1,2} = \frac{1}{2} \left(\frac{\sigma + \beta\sigma + \beta\phi_y + \kappa}{\beta\sigma} \pm \frac{1}{\beta\sigma} \sqrt{(\sigma + \beta\sigma + \beta\phi_y + \kappa)^2 - 4\beta\sigma(\sigma + \phi_y + \kappa\phi_\pi)} \right)$$

$$\lambda_{1,2} = \frac{1}{2\beta\sigma} \left(\sigma + \beta\sigma + \beta\phi_y + \kappa \pm \sqrt{(\sigma + \beta\sigma + \beta\phi_y + \kappa)^2 - 4\beta\sigma(\sigma + \phi_y + \kappa\phi_\pi)} \right)$$

Then:

$$\lambda_1 = \frac{1}{2\beta\sigma} \left(\sigma + \beta\sigma + \beta\phi_y + \kappa - \sqrt{(\sigma + \beta\sigma + \beta\phi_y + \kappa)^2 - 4\beta\sigma(\sigma + \phi_y + \kappa\phi_\pi)} \right)$$

$$\lambda_2 = \frac{1}{2\beta\sigma} \left(\sigma + \beta\sigma + \beta\phi_y + \kappa + \sqrt{(\sigma + \beta\sigma + \beta\phi_y + \kappa)^2 - 4\beta\sigma(\sigma + \phi_y + \kappa\phi_\pi)} \right)$$

With the eigenvalues we can find the eigenvectors. Evaluating $(\Omega - \lambda I)\mathbf{v} = \mathbf{0}$ at $\lambda = \lambda_1$ we get the right-eigenvector associated to λ_1 , denoted by $\mathbf{v}^1$. We have:

$$\begin{bmatrix} \frac{1}{\beta} - \lambda_1 & -\frac{\kappa}{\beta} \\ \frac{\beta\phi_\pi - 1}{\beta\sigma} & \frac{\beta(\sigma + \phi_y) + \kappa}{\beta\sigma} - \lambda_1 \end{bmatrix} \begin{bmatrix} v_1^1 \\ v_2^1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Then:

$$\left(\frac{1}{\beta} - \lambda_1 \right) v_1^1 - \frac{\kappa}{\beta} v_2^1 = 0$$

$$\frac{\beta\phi_\pi - 1}{\beta\sigma} v_1^1 + \left(\frac{\beta(\sigma + \phi_y) + \kappa}{\beta\sigma} - \lambda_1 \right) v_2^1 = 0$$

⁷See Chapter 21 in Sydsæter, Strøm & Berck (2005) for a concise exposition.

By linear dependence (recall $\det(\Omega - \lambda I) = 0$) we can use any one of the two equations above. From the first one we get: $(\frac{1}{\beta} - \lambda_1)v_1^1 = \frac{\kappa}{\beta}v_2^1 \Rightarrow v_2^1 = \frac{\beta}{\kappa}(\frac{1}{\beta} - \lambda_1)v_1^1 \Rightarrow$

$$\mathbf{v}^1 = \begin{bmatrix} v_1^1 \\ \frac{\beta}{\kappa}(\frac{1}{\beta} - \lambda_1)v_1^1 \end{bmatrix}$$

Normalizing $v_1^1 = 1$ we get:

$$\mathbf{v}^1 = \begin{bmatrix} 1 \\ \frac{1}{\kappa}(1 - \beta\lambda_1) \end{bmatrix}$$

Analogously, for λ_2 we get:

$$\mathbf{v}^2 = \begin{bmatrix} 1 \\ \frac{1}{\kappa}(1 - \beta\lambda_2) \end{bmatrix}$$

Now we can construct the matrix of eigenvectors as follows:

$$\Gamma = \begin{bmatrix} 1 & 1 \\ \frac{1}{\kappa}(1 - \beta\lambda_1) & \frac{1}{\kappa}(1 - \beta\lambda_2) \end{bmatrix}$$

It is not difficult, but tedious, to show that $\Gamma\Lambda\Gamma^{-1} = \Omega$.⁸

Finally, we want to say something about the sign and size of the eigenvalues. We first notice that the characteristic polynomial can be written as follows:

$$\begin{aligned} P(\lambda) &= \lambda^2 - \frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma}\lambda + \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma} \\ &= \lambda^2 - \text{tr}(\Omega)\lambda + \det(\Omega) \\ &= \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2 \\ &= (\lambda - \lambda_1)(\lambda - \lambda_2) \end{aligned}$$

where we have used $\lambda_1 + \lambda_2 = \text{tr}(\Omega)$ and $\lambda_1\lambda_2 = \det(\Omega)$.

Now we notice that

$$\begin{aligned} \lambda_1 + \lambda_2 &= \frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma} > 1 \\ \lambda_1\lambda_2 &= \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma} > 1 \end{aligned}$$

because in both cases the numerator is higher than σ while the denominator is strictly smaller than σ .

From $\lambda_1\lambda_2 > 1 > 0$ we learn that both eigenvalues must have the same sign. Then, $\lambda_1 + \lambda_2 > 1$ implies that both eigenvalues must be positive. Then, $\lambda_1\lambda_2 > 1$ implies that either $\lambda_1 > 1$ or $\lambda_2 > 1$ (or both). Now we evaluate $P(\lambda)$ at $\lambda = 1$ to get: $P(1) = (1 - \lambda_1)(1 - \lambda_2)$. Since either $\lambda_1 > 1$ or $\lambda_2 > 1$, it follows that $P(1) > 0 \Leftrightarrow \lambda_1 > 1$ and $\lambda_2 > 1$. But $P(1) = 1 - \frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma} + \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma}$. Then, $P(1) > 0 \Leftrightarrow 1 - \frac{\sigma + \beta(\sigma + \phi_y) + \kappa}{\beta\sigma} + \frac{\sigma + \phi_y + \kappa\phi_\pi}{\beta\sigma} > 0$. Then: $P(1) > 0 \Leftrightarrow \kappa(\phi_\pi - 1) + (1 - \beta)\phi_y > 0$. Finally, we conclude:

$$\lambda_1 > 1 \text{ and } \lambda_2 > 1 \Leftrightarrow \kappa(\phi_\pi - 1) + (1 - \beta)\phi_y > 0$$

which gives a necessary and sufficient condition for the existence of a unique non-explosive solution. Notice that a sufficient (but not necessary) condition is $\phi_\pi > 1$.

We can give some intuition for the expression above. Consider a situation with constant inflation and output gap. From the Taylor Rule we get: $\frac{\partial i}{\partial \pi} = \phi_\pi + \phi_y \frac{\partial \tilde{y}}{\partial \pi}$. But the Phillips Curve gives: $\pi = \beta\pi + \kappa\tilde{y} \Rightarrow \tilde{y} = \frac{1-\beta}{\kappa}\pi \Rightarrow \frac{\partial \tilde{y}}{\partial \pi} = \frac{1-\beta}{\kappa}$. Substituting into the expression for $\frac{\partial i}{\partial \pi}$ we get: $\frac{\partial i}{\partial \pi} = \phi_\pi + \phi_y \frac{1-\beta}{\kappa}$. Then, $\frac{\partial i}{\partial \pi} > 1 \Leftrightarrow \phi_\pi + \phi_y \frac{1-\beta}{\kappa} > 1$. But $\phi_\pi + \phi_y \frac{1-\beta}{\kappa} > 1 \Leftrightarrow \kappa(\phi_\pi - 1) + (1 - \beta)\phi_y > 0$. Then:

$$\frac{\partial i}{\partial \pi} > 1 \Leftrightarrow \kappa(\phi_\pi - 1) + (1 - \beta)\phi_y > 0$$

Hence, the nominal interest rate must increase more than one-for-one with the inflation rate. In other words, the *real* interest rate must increase in response to inflation.

⁸ Notice that $\det(\Gamma) = \frac{\beta}{\kappa}(\lambda_1 - \lambda_2)$. Hence, for Γ to be invertible we need $\lambda_1 \neq \lambda_2$.

References

Blanchard, O. and Kahn, C. (1980). "The Solution of Linear Difference Models under Rational Expectations," *Econometrica*, Vol. 48 No. 5 (July), pp. 1305-1311.

Galí, Jordi (2008). *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework*, Princeton University Press.

Sydsæter, K., Strøm, A. and Berck, P. (2005). *Economists' Mathematical Manual*, Fourth Edition, Springer.